SHEA'S TECH | PROFESSIONAL TRAINING SERIES

RHCSA Course

From Learn Linux For Work

Zero to Red Hat Certified System Administrator (EX200)
EX200 · Red Hat Enterprise Linux 10

rhcsa.learnlinuxforwork.com • learnlinuxforwork.com

12 weeks • 12 lab guides • 10 exam domains
Prepared for Normal Broke People
Edition: August 2026

Contents

What's inside this eBook

Every lab guide lives online

This eBook mirrors the course. The twelve full lab guides — with the commands, verification steps, and break/fix drills — are on the site at rhcsa.learnlinuxforwork.com, and they print cleanly if you prefer paper.

01 How This Course Works

Twelve weeks from first shell prompt to a passing score

The RHCSA is not a multiple-choice exam. You are given a live Red Hat Enterprise Linux system, a list of tasks, and a few hours. Nothing is graded on how you did it — only on whether the machine ends up in the required state, and whether it stays that way after a reboot. That single fact should shape everything about how you study.

So this course is built around doing, not reading. Every week has a focus, a short list of tasks, and a **full lab guide** with the actual commands, the verification step, and a break/fix drill that forces you to recover from something you broke on purpose. If you finish a week without having typed anything, you haven't finished the week.

The current exam is **EX200 on Red Hat Enterprise Linux 10**. That matters — RHEL 10 added Flatpak package management to the objectives, and some older topics have moved on. Everything here targets the RHEL 10 objective list.

Where the hours go

- **70%** hands-on labs on a machine you can break and rebuild
- **20%** reading the man pages and system docs — the only reference you get in the exam
- **10%** review — write down every command that didn't work the first time

Pacing options

Track	Duration	Hrs/wk	How it runs
Standard	12 weeks	8–10	One week per module. Realistic for a working adult starting near zero.
Intensive	6 weeks	16–20	Two modules per calendar week. Best if you have prior Linux exposure or are job-searching full time.
Extended	18 weeks	5–6	Stretch each module by half. Better than quitting because the pace was unrealistic.

The rule that fails more candidates than any other

Configurations must persist after reboot without intervention. That is written into the objectives. A mount that works until you reboot is worth zero. A service that runs but isn't enabled is worth zero. Build the habit now: after every single lab in this course, run `reboot` and confirm your work survived. If you only take one thing from this page, take that.

02 What the RHCSA Actually Is

Format, conditions, and what the graders care about

Knowing the shape of the exam changes how you practise. This is a performance-based, hands-on exam — you sit at a real RHEL 10 system and complete real administration tasks. There are no questions to answer and no partial credit for knowing the theory.

Exam code	EX200
Based on	Red Hat Enterprise Linux 10
Format	Performance-based — real tasks on a live system
Internet access	None during the exam
Notes or books	Not permitted, in any form
Documentation	The docs that ship with the product are available — man pages, info, /usr/share/doc
Persistence	Every configuration must survive a reboot with no intervention
Leads to	Prerequisite for RHCE (Enterprise Linux and Ansible tracks)

What that means for how you study

- **Learn the man pages, not the tutorials.** They are the only reference you get. Practise finding things with `man -k`, `man 5 fstab`, and `/usr/share/doc` until it is second nature.
- **Speed matters, but correctness matters more.** A half-finished task scores nothing. Finish and verify each task before moving to the next.
- **Reboot constantly while practising.** Make it a reflex, not a checklist item.
- **Never memorise a GUI path.** Everything in this course is done at the command line, because that is what is reliably available.
- **There is no partial credit for approach.** If `vim` and a text editor both get the file right, both are correct. Use whatever you are fastest with.

> **Read the objectives yourself — from the source**
> Red Hat publishes the full EX200 objective list, and it is the only authoritative statement of what is testable. Everything in this course maps to it. Bookmark it and re-read it in Week 11: redhat.com — EX200 exam objectives.

03 Build Your Home Lab

Two virtual machines, a free hypervisor, and a Red Hat developer account — that's the whole setup

You need at least two RHEL 10 systems you are willing to destroy: one to work on, one to practise remote access, file transfer, and NFS against. Virtual machines are ideal — they snapshot, they roll back, and breaking one costs you nothing.

A laptop you already own plus a free hypervisor and a free Red Hat developer account is the entire minimum investment. Everything below assumes that setup.

Start here — the three-step path

- Create a free Red Hat account and activate the **Developer Subscription for Individuals**. It covers up to **16 systems** at no cost.
- Install **RHEL 10** on real hardware, or two VMs on whatever machine you already own. Rocky Linux 10 is the free rebuild if you would rather not register.
- Snapshot both machines. Then break them on purpose, every single week, and put them back.

Don't overbuy

Two virtual machines on a laptop you already own costs nothing and is enough for the entire exam. Buy hardware only if your current machine can't spare 8 GB of RAM. There is no lab in this course that needs a server, a rack, or a homelab YouTube shopping list.

03 Pick a hypervisor and size the VMs

Everything in this course runs on two virtual machines. Any of these will do it — pick the one that matches your hardware and stop researching....

Hypervisor	Why you'd pick it	Cost
Oracle VirtualBox **virtualbox.org**	**The default choice for this course.** Free, open source, and runs on Linux, macOS (Intel), and other desktops. Snapshots are immediate and easy to roll back, which matters enormously here — you will break these machines every week. Use the built-in *Host-only Adapter* for the private lab network.	**Free**
VMware Workstation Pro **Broadcom — support.broadcom.com**	Now free for personal use under Broadcom. Noticeably faster disk and network performance than VirtualBox on most hardware, and its snapshot tree is better if you like branching. Fusion is the equivalent on Mac. Use a *Host-only* or *Custom (VMnet)* network for the lab segment.	**Free for personal use**
KVM / virt-manager **libvirt on a Linux host**	If your daily driver is already Linux, this is the fastest and lightest option — it is the same virtualisation stack RHEL itself uses, so what you learn transfers. Install with `dnf install -y virt-manager libvirt qemu-kvm` and use the default NAT network plus an isolated network for the lab.	**Free**
Proxmox VE **proxmox.com**	Only worth it if you have a spare machine to dedicate. Runs on bare metal, managed from a browser, and lets you leave the lab running permanently without tying up your laptop. Overkill for RHCSA alone, but it pays off if you continue toward RHCE or the DevOps roadmap.	**Free**
UTM **mac.getutm.app**	The practical option on an Apple Silicon Mac, where VirtualBox does not run. It emulates or virtualises aarch64 guests — RHEL 10 ships an aarch64 ISO, so this works. Slower than x86 on x86, but perfectly usable for everything in this course.	**Free**
Hands-on labs in the browser **learnlinuxforwork.com**	No hypervisor, no download, no spare RAM. Guided hands-on Linux labs that run in a browser tab — useful when you are away from your own machines, or if your laptop simply cannot host VMs. Pair it with Red Hat's free interactive labs.	**Free / hosted**

Recommended VM configuration

- **Host operating system:** any of these hypervisors run on Ubuntu, Fedora, RHEL, Rocky, or another Linux desktop. **Ubuntu is the easiest host to get running** on a used laptop or a home server — install it, add VirtualBox or VMware Workstation, and build your RHEL and Rocky guests on top. The host distro has no bearing on the exam; only the guests matter.
- **servera (primary):** 2 vCPU, 4 GB RAM, 20 GB system disk, **plus a second empty 10 GB disk**. That spare disk is what you partition in Week 8 and turn into LVM in Week 9 — add it at build time and forget about it.
- **serverb (secondary):** 1 vCPU, 2 GB RAM, 20 GB disk. It only ever acts as an SSH target, a file-transfer destination, and an NFS server.
- **Networking:** give both machines two adapters — one NAT adapter so they can reach the internet for package installs, and one *host-only / internal* adapter on a private subnet so they can reach each other at fixed addresses. Week 7 configures static addressing on that second adapter.
- **Firmware:** enable EFI if your hypervisor offers it. RHEL 10 installs happily either way, but EFI matches what you are most likely to meet in the real world.

- **Install type:** choose *Server* or *Minimal Install*, never Workstation. You want to live at a text console, because that is where the exam happens.
- **Snapshots:** take one named `clean` the moment each install finishes, and another named `networked` once Week 7 is done. Roll back to `clean` before every mock exam.
- **Host requirements:** 8 GB of RAM on the host is enough to run both machines at once. With 16 GB you will not notice them at all.

03 Get the operating system

RHEL and Rocky are what you practise the exam on — same commands, same behaviour. Ubuntu has a different job here: it makes a fine host for the...

Operating system	What it is and when to use it	Cost
Red Hat Enterprise Linux 10 **Developer Edition — developers.redhat.com**	The real exam platform. The **Developer Subscription for Individuals** is free and covers up to **16 physical or virtual systems**. Create an account, activate the subscription, download the boot or DVD ISO. This is the one to use. Read the FAQ for the terms.	**Free · 16 systems**
Rocky Linux 10 **rockylinux.org**	A community rebuild of RHEL 10 — same commands, same behaviour, no account required. Use it for extra practice nodes, or if you want to start tonight without registering anything. One caveat: subscription-manager and Red Hat CDN tasks only exist on real RHEL.	**Free · unlimited**
Ubuntu **Canonical — ubuntu.com**	Ubuntu is not RHEL-compatible, so do not use it as your *exam practice* guest — the package manager, SELinux behaviour, and defaults all differ. But it is an excellent **host** operating system: install Ubuntu on the laptop or server, run VirtualBox, VMware Workstation, or KVM on top of it, and boot your RHEL and Rocky guests inside. It is also the distro you are most likely to meet at work, so a third VM running it is time well spent.	**Free · host or guest**

Getting it installed

- Download the ISO from developers.redhat.com (RHEL) or rockylinux.org (Rocky).
- Write it to a USB stick with `dd` or balenaEtcher if you are installing to bare metal.
- Choose **Server** or **Minimal Install** at the package-selection screen — not Workstation. You want to live at a text console, because that is where the exam happens.
- Give each VM 2 CPUs, 2–4 GB RAM, and a 20 GB disk, plus **a second empty 10 GB disk**. You will need that spare disk from Week 8 onward for partitioning and LVM. Add it now and forget about it.
- Take a snapshot the moment the install finishes and you have logged in once. Name it `clean`. You will roll back to it more often than you expect.

03 Set up the two-machine topology

A handful of objectives — SSH, key-based authentication, secure file transfer, NFS, autofs — need a second system. Build both now so you are never...

Machine	Role	Suggested name
Primary	Where you do almost everything. Partitioning, LVM, SELinux, services, users, scripting. This is the one you break.	**servera**
Secondary	The remote end. SSH target, scp/rsync destination, NFS server for the storage weeks. Lighter — 1 CPU and 1 GB RAM is fine.	**serverb**

Wire them together once, at the start

- Put both VMs on the same host-only or internal network so they can reach each other by IP.
- Add both to `/etc/hosts` on each machine so `ssh serverb` works without DNS. You will configure real name resolution in Week 7.
- Confirm you can `ssh` both directions before Week 1 ends. Nearly every later lab assumes it.
- Snapshot both again once networking works. Name it `networked`.

03 Build the local lab with one script

Two ready-to-run scripts live in the scripts/ folder of the course repo. Both create the same two machines — servera and serverb — with the spare...

Script	What it does	Where it runs
build-local-lab.sh **VirtualBox / VMware**	Creates `/rhcsa-labs`, tells you exactly which ISO to download into it, then builds both VMs in **Oracle VirtualBox** or **VMware Workstation Pro** — correct RAM, CPU, EFI firmware, NAT plus host-only networking, and the spare disk already attached. Run `--destroy` to wipe and start over.	**Your Linux laptop**
build-podman-lab.sh **Podman**	Two **Rocky Linux 10 Podman containers** with systemd running inside, for machines with no spare RAM at all. Good for Weeks 1–4, 10, and 11. **It cannot do Weeks 5–9** — containers share the host kernel, so storage, LVM, boot, and firewalld are off the table. The script says so plainly rather than pretending otherwise.	**Any machine**

Getting the ISO onto a Linux laptop

- The script creates `/rhcsa-labs` and takes ownership of it for your user. Put the ISO there and it is found automatically.
- **RHEL 10** — sign in and download from developers.redhat.com/products/rhel/download. It requires a free account, which is also what gives you the 16-system subscription.
- **Rocky 10** — no account needed, so you can pull it straight from the terminal:
  ```
  cd /rhcsa-labs && curl -LO https://download.rockylinux.org/pub/rocky/10/isos/x86_64/Rocky-10-latest-x86_64-dvd.iso
  ```
- Verify the checksum before you install. Every one of those pages publishes one, and a corrupted ISO wastes an afternoon.

03 Used laptops (cheapest, recommended to start)

Only if your current machine can't spare the memory. Off-lease corporate laptops are the best value in computing — look for an Intel i5/i7 or...

Seller	What to expect	Condition
eBay — Renew4me	Cleaned, tested, refurbished machines. Pay a little more, get something that works out of the box. Best first stop if this is your only lab machine.	**Refurbished**
eBay — NurTech	Cheapest and roughest. Expect cosmetic damage and possible self-repairs (battery, keyboard, screen). Great value if you're comfortable with a screwdriver.	**Budget / as-is**
Back Market	Marketplace of vetted refurbishers with a 1-year warranty and a 30-day return window — the safest used purchase if you have never bought refurbished before. That link is pre-filtered to laptops under $300.	**Warrantied · under $300**
Refurb.me	Price tracker and aggregator for refurbished Macs, including Apple's own certified refurbished stock. Set an alert and wait for a deal rather than buying on the day you start looking.	**Deal tracker**
Mac of All Trades	Long-running used Mac dealer with graded conditions and warranties. Older Intel MacBook Pros land cheap here and make excellent Linux laptops.	**Used Macs**

Yes — you can wipe a used MacBook and install Linux on it

- A used Mac is just a laptop. Wipe macOS entirely, install RHEL, Rocky, or Ubuntu, and you have a lab machine with a great screen, keyboard, and build quality for the price of a beat-up plastic PC.
- **Intel Macs (roughly 2012–2020) are the safe buy.** Linux installs and runs on them the same way it does on any other x86 laptop — boot from USB, wipe the disk, done. These are the cheapest good laptops on the used market right now precisely because macOS has dropped support for many of them.
- **Apple Silicon Macs (M1 and later) are the harder path.** Linux on them means Asahi Linux, which is impressive but still incomplete, and RHEL and Rocky do not target that hardware at all. For an RHCSA lab, buy Intel. If you already own an Apple Silicon Mac, use UTM with the aarch64 RHEL ISO instead of wiping the machine.
- Hold `Option` at boot to pick the USB installer. On a T2-chip Mac (2018–2020) you must first disable Secure Boot in Recovery → Startup Security Utility, or the installer will not boot.
- Check the battery cycle count and the keyboard before you buy — the 2016–2019 butterfly keyboards were a known failure point.

03 Buy a new laptop built for Linux

If you'd rather not gamble on used hardware, these vendors ship Linux preinstalled and tested. More expensive, zero driver hunting.

Vendor	Notes	Based in
Laptop with Linux	Clevo hardware, wide config options	Netherlands
System76	Clevo hardware; ships with Pop!_OS or Ubuntu	USA
Framework	Modular and repairable — highly customizable	USA
Purism	Privacy and security focused, hardware kill switches	USA
StarLabs	Compact Linux-first laptops, coreboot firmware	UK / USA
NovaCustom	Clevo hardware, coreboot options	Netherlands
Tuxedo Computers	Built in Germany, own TUXEDO OS	Germany
Slimbook	Assembled in Spain, thin and light	Spain
NitroKey	Security-hardened hardware and keys	Germany
Eurocom	Heavy-duty mobile workstations	Canada
Lenovo ThinkPad	Certified Linux ThinkPads, easiest to buy	Global

03 Add a home server later (optional)

Not needed for the RHCSA — two VMs on a laptop cover every objective. Worth it only if you plan to continue toward RHCE, Kubernetes, or the DevOps...

Vendor	Notes	Site
The Server Store	Wide inventory, configure-to-order	**theserverstore.com**
Server Monkey	Refurb servers, parts, warranties	**servermonkey.com**
New Server Life (NSL)	Onsite colocation: Miami, Dallas, SF, Warsaw	**newserverlife.com**
TechMikeNY on eBay	Well-reviewed eBay storefront	**ebay.com/str/techmikeny**
TechMikeNY (direct)	Buy direct, skip eBay fees	**techmikeny.com**
SaveMyServer	Refurbished enterprise gear	**savemyserver.com**
UsedServers	Used servers, storage, networking	**usedservers.com**
PC Server & Parts	Refurbished servers and spare parts	**pcserverandparts.com**

What to run on it

- Target a Dell PowerEdge R630/R730 or HP DL360/DL380: dual CPUs, 64–128 GB RAM, a few hundred dollars. They are loud — a closet or garage is ideal.
- Proxmox VE is the home-lab default: Debian-based, web UI, KVM VMs plus LXC containers, snapshots, and clustering.
- OpenNebula is the alternative if you want something closer to a real private cloud. Steeper curve, better multi-node story.

03 Networking gear (optional)

Entirely optional for the RHCSA, which only asks for addressing, name resolution, and firewalld. Useful if you want to push toward hybrid-cloud...

Hardware	What it's for
Protectli **protectli.com**	Fanless firewall appliances. Run pfSense, OPNsense, or VyOS and build a real perimeter around the lab.
CCNA Lab Racks **eBay — GeoNetwork US**	Prebuilt racks of Cisco switches and routers — real IOS, real cabling, real console ports.

04 Self-Managed Cloud Hosted Labs

The same two machines on a provider you rent by the hour — for when the laptop can't spare the RAM

If your machine cannot host two virtual machines, rent them instead. A small instance pair costs roughly $10–25 per month if you leave it running, and far less if you destroy it between study sessions — which you should.

`scripts/build-cloud-lab.sh` builds the identical `servera` / `serverb` pair on any of the four providers below, including the spare disk for the storage weeks, firewall rules that let the two hosts talk to each other, and SSH locked to your own address. Every provider block has a matching `--destroy`.

One genuine upside: rebuilding the lab from nothing every week is *itself* RHCSA practice. You will install, partition, and configure far more often than someone whose VMs just sit there.

Provider	Why you'd pick it	Command
Google Cloud **console.cloud.google.com**	Rocky Linux 10 images are published in the `rocky-linux-cloud` project, so no marketplace hunting. New accounts get free credit, which can cover most of a 12-week run. Needs the gcloud SDK and a `gcloud init`.	`./build-cloud-lab.sh gcp`
DigitalOcean **cloud.digitalocean.com**	The simplest pricing of the four and a Rocky 10 image already in the catalogue. Block volumes attach cleanly, which makes the LVM week straightforward. Needs doctl.	`./build-cloud-lab.sh do`
Amazon Web Services **console.aws.amazon.com**	Rocky 10 AMIs are available, and the script finds the newest one for your region automatically. Note the free tier does *not* cover t3.medium — set a billing alarm on day one. Needs the AWS CLI.	`./build-cloud-lab.sh aws`
Vultr **my.vultr.com**	Cheap hourly instances and inexpensive block storage for the LVM labs. Often the lowest total cost if you are disciplined about destroying the lab. Needs vultr-cli and an API key.	`./build-cloud-lab.sh vultr`

Before you spend anything

- **Set a billing alert on day one.** Every provider offers one free, and every provider will happily bill you for an instance you forgot about.
- Generate an SSH key first — the script requires one: `ssh-keygen -t ed25519 -C 'shea@rhcsa-lab'`
- **Destroy the lab when you finish for the week:** `./scripts/build-cloud-lab.sh <provider> --destroy`. Stopping an instance still bills you for its disk.
- A cloud lab cannot practise everything. Console-level work — interrupting the boot with `rd.break`, resetting a lost root password, editing the bootloader — needs a hypervisor where you control the virtual console. Do Week 5 locally even if everything else is hosted.
- If you would rather not manage any of it, the browser-based hands-on labs at learnlinuxforwork.com and Red Hat's interactive labs need no setup and no credit card.

04 The Certification Ladder

Two optional confidence builders before you spend $500 on the real thing

RHCSA proves you can administer a Red Hat system. It is expensive and unforgiving — one sitting, performance-based, pass or fail. If you have never taken a certification exam, walking into that as your first one is a lot of pressure on a big cheque.

Three cheaper exams sit below it, in increasing order of difficulty. None is required. Each is worth considering if the cost of failing the RHCSA cold would set you back — and you can stop at any rung.

1

LPI Linux Essentials

010-160 | Entry level · optional starting point
Optional — after Week 3 | ~$120

The gentlest on-ramp there is. Vendor-neutral, single exam, and aimed at people who are genuinely new to Linux rather than at working administrators. If you have never sat a proctored exam or you are still shaky on the command line after Week 3, this is a cheap, low-pressure way to prove to yourself that you can do this. Skip it if you already work in IT.

2

CompTIA Linux+ (XK0-006, v8)

XK0-006 | Foundational · optional confidence builder
Optional — after Week 6 | ~$369

Vendor-neutral and broader than RHCSA — it covers scripting, security, and troubleshooting across distributions, and the current version leans into automation and cloud-adjacent topics. It is partly multiple-choice, so it is a gentler introduction to sitting a timed, proctored exam. If you pass it, you will walk into the RHCSA knowing you can handle the room.

3

Linux Foundation Certified Systems Administrator

LFCS | Associate · optional confidence builder
Optional — after Week 9 | ~$395

The closest thing to an RHCSA dress rehearsal. It is also performance-based — a real terminal, real tasks, a timer — but it is distro-flexible and slightly more forgiving. The overlap with RHCSA is large: users, permissions, storage, networking, services, and scripting. Sitting LFCS first means the RHCSA format holds no surprises. Linux Foundation exam registrations include access to the killer.sh simulator, which is the closest you will get to sitting the real thing beforehand.

4

Red Hat Certified System Administrator

EX200 | The goal · performance-based on RHEL 10
End of Week 12 | ~$500

The one employers ask for by name in Red Hat shops, and a hard prerequisite for RHCE. Entirely hands-on, entirely on RHEL 10, and everything must survive a reboot.

If you work for a U.S. government contractor: DoD 8140

DoD Directive 8140 — which replaced DoD 8570.01-M — governs cyber workforce qualification across the Department of Defense and its contractors. If you administer DoD systems, the certifications you hold are not a nice-to-have. They decide whether you are allowed to hold an account at all.

The part that matters most to a Linux administrator is **privileged user access (PUA)**. A privileged account — root, sudo, anything that can change a system's security posture — generally requires two things, not one:

1. **A baseline security certification** for the work role and level. CompTIA Security+ is the long-standing entry point and the one most contracts name by default. CompTIA CySA+ covers analyst and higher-tier roles, and is the usual next step.
2. **A computing environment (CE) certification** for the platform you actually administer. **This is where the RHCSA comes in.** For Red Hat Enterprise Linux estates — which is most of the federal Linux footprint — the RHCSA is the standard CE credential.

The pairing that unlocks a privileged account

RHCSA + Security+ is the combination you will see on the majority of federal Linux system administrator job postings. The RHCSA proves you can run the platform; Security+ satisfies the baseline requirement. Add **CySA+** when you move toward security operations or a higher IAT tier. Many contractors will not onboard you to a privileged role until both are in hand, and quite a few will pay for the second one once you have the first.

Verify before you spend money on this

DoD 8140 moved from a fixed certification matrix to a **work-role-based qualification framework**, and the approved certification lists are revised periodically. What your specific contract, agency, and work role require may differ from the general pattern above. Check the current lists on the DoD Cyber Exchange and confirm with your facility security officer or contract lead *before* booking an exam on the strength of this section.

Skip both if you're confident and short on money

Linux+ and LFCS together cost more than the RHCSA itself. They are confidence builders, not prerequisites — nothing about them makes your RHCSA more likely to pass beyond the practice and the calmer nerves. If you have sat proctored exams before and money is tight, go straight at EX200 and put the savings toward a retake voucher instead. After RHCSA, the natural next step is RHCE (EX294), which is Ansible automation.

05 Exam Objective Coverage Map

All ten of Red Hat's EX200 objective domains, mapped to the week that covers them

Red Hat groups the EX200 objectives into ten categories. Every one is covered, and nothing here is filler. If you can do everything in the right-hand column without notes, you are ready.

Red Hat objective domain	What you'll actually be able to do	Week
Understand and use essential tools	Shell syntax, redirection and pipes, grep with regular expressions, SSH to remote systems, switching users, tar/gzip/bzip2 archives, creating and editing files, hard and soft links, ugo/rwx permissions, and finding answers in man, info, and /usr/share/doc	**Wk 1**
Manage users and groups	Create, modify, and delete local users and groups, set and age passwords, manage supplementary group membership, and configure privileged access with sudo	**Wk 2**
Manage software	Configure RPM repository access, install and remove RPM packages, configure Flatpak remotes, and install and remove Flatpak applications	**Wk 3**
Operate running systems — processes	Identify CPU and memory hogs, kill processes, adjust scheduling priority, manage tuned profiles, and schedule work with at, cron, and systemd timers	**Wk 4**
Operate running systems — boot and services	Boot, reboot, and shut down cleanly, boot into different targets, interrupt the boot process to recover a system, set the default target, start/stop/enable services, and modify the bootloader	**Wk 5**
Operate running systems — logs and time	Locate and read system logs and journals, make the journal persistent, and configure time service clients	**Wk 6**
Manage basic networking	Configure IPv4 and IPv6 addresses, set hostname resolution, make network services start at boot, and restrict access with firewalld and firewall-cmd	**Wk 7**
Configure local storage	List, create, and delete GPT partitions, add swap non-destructively, and mount filesystems at boot by UUID or label	**Wk 8**
Create and configure file systems	Create and mount VFAT, ext4, and XFS filesystems, build and extend LVM volumes, mount NFS shares, configure autofs, and diagnose permission problems	**Wk 9**
Manage security	Set SELinux enforcing and permissive modes, read and restore file contexts, manage port labels and booleans, configure key-based SSH authentication, and set default permissions with umask	**Wk 10**
Create simple shell scripts	Conditional execution with if and test, loops to process files and input, positional parameters, and using the output of commands inside a script	**Wk 11**
Putting it together	A full timed mock exam under real conditions, remediation of everything you missed, and exam-day logistics	**Wk 12**

06 The 12-Week Plan

Twelve modules. Every task is something you do, not something you read.

1 Essential Tools and the Shell

Understand and use essential tools · 8–10 hrs

The commands you will use in every other week. Get fast here and everything downstream gets easier.

- Navigate the filesystem hierarchy and explain what lives in /etc, /var, /usr, and /home
- Use redirection and pipes fluently: >, >>, |, 2>, 2>&1, and tee
- Search text with grep and basic regular expressions — anchors, character classes, and quantifiers
- Create, copy, move, and delete files and directories, including recursively and safely
- Create hard links and symbolic links, and explain when each breaks
- Read and set ugo/rwx permissions in both symbolic and octal form
- Archive and compress with tar, gzip, and bzip2, and unpack all three
- Connect to your second machine over SSH and switch users with su and sudo
- Find answers using man, man -k, info, and /usr/share/doc — no search engine

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-01.html

2 Users, Groups, and Privileged Access

Manage users and groups · 8–10 hrs

Small domain, heavily tested, and easy marks — as long as you also get the sudo configuration right.

- Create, modify, and delete local user accounts with useradd, usermod, and userdel
- Read /etc/passwd, /etc/shadow, and /etc/group and explain every field
- Set and change passwords, and configure password aging with chage
- Create and delete groups, and manage primary versus supplementary membership
- Configure sudo access for a user and for a group using a drop-in file in /etc/sudoers.d
- Set up an account that must change its password at first login
- Configure defaults for new accounts with /etc/skel and useradd -D

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-02.html

3 Software: RPM, Repositories, and Flatpak

Manage software · 8–10 hrs

RHEL 10 added Flatpak to the objectives. Most study material predates that — do not skip it.

- Install, remove, and query packages with dnf and rpm
- Configure access to an RPM repository by writing a .repo file in /etc/yum.repos.d
- Install a package from a local file, from a configured repository, and from the Red Hat CDN
- List enabled repositories, and enable or disable one temporarily and permanently
- Find which package owns a given file, and list the files a package installed
- Configure access to a Flatpak remote
- Install, list, run, and remove a Flatpak application
- Explain the difference between an RPM and a Flatpak, and when each is appropriate

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-03.html

4 Processes, Tuning, and Scheduled Tasks

Operate running systems · 8–10 hrs

Finding the runaway process, changing its priority, and making work happen on a schedule.

- Identify CPU and memory intensive processes with ps, top, and uptime
- Send signals with kill, killall, and pkill, and explain SIGTERM versus SIGKILL
- Adjust scheduling priority with nice and renice
- List, query, and switch tuned profiles with tuned-adm
- Schedule a one-off job with at
- Schedule recurring jobs with cron, both as root and as a regular user
- Create a systemd timer unit and its matching service unit, and enable it
- Explain when to reach for a timer instead of cron

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-04.html

5 Boot, Targets, Services, and Recovery

Operate running systems · 8–10 hrs

The week that saves you if something goes wrong in the exam. Learn to break the boot and recover it.

- Boot, reboot, and shut down cleanly with systemctl
- List available targets and switch between them at runtime
- Set the default boot target permanently
- Interrupt the boot process at the GRUB menu and boot to an emergency shell
- Reset a forgotten root password from that shell, including the SELinux relabel step
- Start, stop, restart, enable, disable, and check the status of services
- Explain the difference between started and enabled, and why it matters after a reboot
- Modify the bootloader configuration and regenerate it correctly

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-05.html

6 Logs, Journals, and Time

Operate running systems · 8–10 hrs

Where to look when something is broken, and how to make the evidence survive a reboot.

- Read the journal with journalctl and filter by unit, priority, boot, and time range
- Make the systemd journal persistent across reboots and verify it worked
- Locate and interpret traditional log files under /var/log
- Explain the relationship between journald and rsyslog on RHEL
- Configure a time service client with chrony and confirm synchronisation
- Set the system timezone and read the hardware versus system clock
- Securely transfer files between your two machines with scp, sftp, and rsync

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-06.html

7 Networking and firewalld

Manage basic networking · 8–10 hrs

Static addressing that survives a reboot, name resolution, and a firewall that lets the right things through.

- Inspect current configuration with ip addr, ip route, and nmcli
- Configure a static IPv4 address, gateway, and DNS with nmcli and make it persistent
- Configure an IPv6 address on the same interface
- Set the system hostname with hostnamectl
- Configure name resolution using /etc/hosts and /etc/resolv.conf, and test with getent and dig
- Confirm NetworkManager is enabled so networking comes up at boot
- List firewalld zones and services, and inspect the active zone
- Open a service and a specific port permanently, then reload and verify
- Reboot and confirm every one of the above survived

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-07.html

8 Partitions, Filesystems, and Swap

Configure local storage · 10–12 hrs

This is where the second disk you added in the home lab starts earning its keep.

- List block devices and existing partitions with lsblk, blkid, and fdisk -l
- Create a GPT partition table and add partitions on your spare disk
- Delete a partition and explain why the kernel may not notice immediately
- Format partitions as XFS, ext4, and VFAT
- Mount filesystems manually, then persistently via /etc/fstab using UUID
- Mount by label as well, and explain when a label beats a UUID
- Add a swap partition non-destructively and enable it at boot
- Reboot and verify every mount and the swap came back automatically

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-08.html

9 LVM, NFS, and autofs

Create and configure file systems · 10–12 hrs

Logical volumes you can grow, network filesystems you can mount, and permission problems you can diagnose.

- Create physical volumes, a volume group, and logical volumes
- Format and mount a logical volume persistently
- Extend a volume group by adding a physical volume
- Extend a logical volume and grow the filesystem on it, online, without data loss
- Remove a logical volume and a physical volume cleanly
- Export a directory over NFS from your second machine
- Mount that NFS share manually and then persistently
- Configure autofs to mount the share on demand and verify it unmounts when idle
- Diagnose and correct a deliberately broken set of file permissions

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-09.html

10 SELinux, SSH Keys, and Default Permissions

Manage security · 10–12 hrs

The domain that trips up more candidates than any other. Do not set SELinux to permissive and move on.

- Check and set SELinux mode with getenforce and setenforce, and make it persistent
- List file and process contexts with ls -Z and ps -Z
- Change a file context with chcon and explain why it does not survive a relabel
- Set a persistent context rule with semanage fcontext and apply it with restorecon
- Add a non-standard port to an SELinux port label so a service can bind to it
- List booleans with getsebool and flip one persistently with setsebool -P
- Read an SELinux denial in the logs and work out what it is telling you
- Generate an SSH key pair and configure key-based authentication to your second machine
- Set default file permissions with umask for a user and system-wide

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-10.html

11 Shell Scripting

Create simple shell scripts · 8–10 hrs

Only simple scripts are tested. You need conditionals, loops, arguments, and command substitution — nothing exotic.

- Write a script with a proper shebang and make it executable
- Use if, elif, else with test and [ ] to branch on a condition
- Test for file existence, directory existence, string equality, and numeric comparison
- Write a for loop that iterates over a list, a glob, and the lines of a file
- Use a while loop to read input line by line
- Handle positional parameters $1, $2, $@, and $#, and fail cleanly when they are missing
- Capture and use the output of a command with $( )
- Check exit statuses with $? and make your script return a meaningful one
- Re-read the whole EX200 objective list and mark anything still shaky

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-11.html

12 Mock Exam, Remediation, and Exam Day

Putting it together · 10–12 hrs

Sit a full timed mock under real conditions. Then fix what it exposed, and book the exam.

- Roll both lab machines back to a clean snapshot
- Sit a full timed mock exam — no notes, no internet, man pages only
- Reboot at the end and re-check every task; score only what survived
- Write down every objective you missed or had to look up
- Rebuild the labs for each missed objective from scratch, untimed
- Sit a second timed mock later in the week and target a clean pass
- Book the real EX200 while the second mock is still fresh
- Read the exam-day checklist in Section 10 and prepare your ID and environment

Lab guide: rhcsa.learnlinuxforwork.com/lab/week-12.html

07 Lab Guides

One standalone guide per week — real commands, verification steps, and a break/fix drill

Each guide is a self-contained page you can work through at the terminal. Every one follows the same shape: what you're building, the commands, how to verify it, a deliberate break to repair, and a reboot check. They print cleanly if you'd rather work from paper.

#	Lab guide	What you build
1	**Essential Tools and the Shell**	The commands you will use in every other week. Get fast here and everything downstream gets easier.
2	**Users, Groups, and Privileged Access**	Small domain, heavily tested, and easy marks — as long as you also get the sudo configuration right.
3	**Software: RPM, Repositories, and Flatpak**	RHEL 10 added Flatpak to the objectives. Most study material predates that — do not skip it.
4	**Processes, Tuning, and Scheduled Tasks**	Finding the runaway process, changing its priority, and making work happen on a schedule.
5	**Boot, Targets, Services, and Recovery**	The week that saves you if something goes wrong in the exam. Learn to break the boot and recover it.
6	**Logs, Journals, and Time**	Where to look when something is broken, and how to make the evidence survive a reboot.
7	**Networking and firewalld**	Static addressing that survives a reboot, name resolution, and a firewall that lets the right things through.
8	**Partitions, Filesystems, and Swap**	This is where the second disk you added in the home lab starts earning its keep.
9	**LVM, NFS, and autofs**	Logical volumes you can grow, network filesystems you can mount, and permission problems you can diagnose.
10	**SELinux, SSH Keys, and Default Permissions**	The domain that trips up more candidates than any other. Do not set SELinux to permissive and move on.
11	**Shell Scripting**	Only simple scripts are tested. You need conditionals, loops, arguments, and command substitution — nothing exotic.
12	**Mock Exam, Remediation, and Exam Day**	Sit a full timed mock under real conditions. Then fix what it exposed, and book the exam.

How to use a lab guide

Read the whole guide first. Then close it and try the tasks from memory. Open it again only when you're stuck — and when you do, write down what you had to look up. That list is your real study guide for Week 12.

08 Core Resource List

Free and low-cost, and every one of them is linked

Resource	Why it's here
The authoritative source	
EX200 exam objectives	Red Hat's official objective list. The only definitive statement of what is testable. Re-read it in Week 11.
Red Hat product documentation	Free, thorough, written by the vendor. The RHEL 10 system administration guides map closely to the exam domains.
Red Hat Interactive Labs	Free browser-based RHEL labs. Zero setup — good for a quick repetition away from your own machines.
Red Hat Developer	Free developer subscription, downloads, and a large library of articles and cheat sheets.
Getting the operating system	
RHEL 10 Developer Edition	Free for up to 16 systems. This is what you should practise on.
Developer subscription FAQ	The terms of the no-cost individual developer subscription, in plain language.
Rocky Linux 10	Free RHEL 10 rebuild, no account, no limits.
Ubuntu	Not for exam practice — but the easiest Linux to install on the laptop or server that will *host* your RHEL and Rocky virtual machines. Also the distro you are most likely to meet at work.
Hypervisors and home lab	
Oracle VirtualBox	Free, cross-platform, one-click snapshots. The default recommendation for this course.
VMware Workstation Pro	Free for personal use under Broadcom. Faster disk and network than VirtualBox on most hardware.
KVM / virt-manager	If your host is already Linux — the same stack RHEL itself uses, so the skills transfer.
UTM	The practical option on Apple Silicon, where VirtualBox does not run. Use the aarch64 RHEL ISO.
Proxmox VE	Bare-metal hypervisor for a dedicated box. Overkill for RHCSA alone, worth it if you continue.
OpenNebula	Full private-cloud platform. Steeper curve, better multi-node story.
Hardware — used and new	
Back Market	Vetted refurbished laptops with a 1-year warranty, pre-filtered under $300.
eBay — Renew4me	Cleaned and tested refurbished machines.
eBay — NurTech	Cheapest and roughest. Fine if you own a screwdriver.
Refurb.me	Price tracker for refurbished Macs. Set an alert and wait for a deal.
Mac of All Trades	Used Macs with warranties. Wipe an Intel MacBook Pro and install Linux.

Resource	Why it's here
Asahi Linux	Linux on Apple Silicon. Impressive but incomplete — buy Intel for an RHCSA lab.
System76	New Linux-first laptops, Pop!_OS or Ubuntu preinstalled.
Framework	Modular and repairable, highly customizable.
Lenovo ThinkPad	Certified Linux ThinkPads, easiest to buy.
Tuxedo Computers	Built in Germany, ships with its own Linux.
The Server Store	Used enterprise servers if you outgrow a laptop.
TechMikeNY	Refurbished servers, buy direct.
Practice under pressure	
LinuxCert Guru	Hands-on, timed, performance-based RHCSA mock exams. Your Week 12 checkpoint.
LabEx	Guided live-environment Linux labs. Useful for extra repetitions between modules.
KodeKloud	Browser-based Linux and DevOps playgrounds with no setup.
Learn Linux For Work — hands-on labs	Guided browser-based Linux labs. No hypervisor, no download, nothing to install.
Training and community	
Learn Linux For Work	Structured, work-focused Linux training built for people heading into system administration roles.
Doc Linux	Command reference and syntax lookups. Keep it open in a second tab while you lab.
Kubecraft Linux	Community classroom for Linux and Kubernetes — lessons plus people to ask when you're stuck.
The Hood	Community space for questions and support.
r/linuxadmin	Plus r/redhat. Good for sanity checks and other people's exam-day write-ups.
Shea's Tech on YouTube	Video walkthroughs, including what a Red Hat certification exam is actually like.
The confidence-builder exams	
CompTIA Linux+ (XK0-006, v8)	Vendor-neutral, partly multiple-choice. A gentler first proctored exam. Optional.
Linux Foundation — LFCS	Performance-based and distro-flexible. The closest thing to an RHCSA dress rehearsal. Optional.
LPI — Linux Essentials	The gentlest vendor-neutral entry point. One exam, aimed at genuine beginners. A low-pressure first proctored exam before the harder ones.
Linux Foundation training and certification	LFCS, LFCE, and the Kubernetes track (CKA, CKAD, CKS). Performance-based and distro-flexible.
killer.sh	Exam simulators for Linux Foundation certifications. LFCS and Kubernetes exam registrations include simulator sessions — use them before the real sitting.
Scripting and fundamentals	

Resource	Why it's here
Automate the Boring Stuff	Free online. Practical Python for sysadmin-style tasks, useful once you're past Week 11.
Pro Git book	Free online. Not on the exam, but you will want version control for your break/fix log.
OverTheWire Bandit	Free shell puzzles. A fun way to build command-line reflexes in Weeks 1–2.
Google SRE Book	Free online. Where to go once the certification is behind you.
After the RHCSA	
RHCE (EX294)	Ansible automation. The natural next step, and it requires a current RHCSA.
Free DevOps Roadmap	The 54-week Linux-to-AWS DevOps roadmap. RHCSA is its Phase 0 — this course is that phase, in depth.
Ansible documentation	Start here the day after you pass.
Cloud providers for a hosted lab	
Google Cloud	Rocky Linux 10 images are published in the rocky-linux-cloud project. New accounts get free credit. Use scripts/build-cloud-lab.sh gcp.
Google Cloud SDK	Install gcloud, then run 'gcloud init' before using the lab script.
DigitalOcean	Simplest pricing of the four and a Rocky 10 image in the catalogue. Use scripts/build-cloud-lab.sh do.
doctl (DigitalOcean CLI)	Needed by the cloud lab script for DigitalOcean.
AWS EC2	Rocky 10 AMIs are in the Marketplace. Free tier does not cover t3.medium, so watch the bill. Use scripts/build-cloud-lab.sh aws.
AWS CLI	Needed by the cloud lab script for AWS.
Vultr	Cheap hourly instances and block storage for the LVM labs. Use scripts/build-cloud-lab.sh vultr.
vultr-cli	Needed by the cloud lab script for Vultr.
Podman	Rootless containers. scripts/build-podman-lab.sh builds a container lab for the weeks that don't need a kernel of their own.
U.S. government contracting	
DoD 8140 — DoD Cyber Exchange	The authoritative source for cyber workforce qualification requirements, work roles, and currently approved certifications. Check it before planning a cert path around a contract.
CompTIA Security+	The baseline security certification most federal contracts name. Pairs with RHCSA for privileged user access on Linux systems.
CompTIA CySA+	Analyst-tier certification, the common next step above Security+ for security operations work roles.
Red Hat certification directory	Where an employer verifies your RHCSA is current. Red Hat certifications expire — usually after three years.

09 Estimated Costs

What this actually costs, with the optional parts marked

Item	Estimate
RHEL 10 Developer Edition (16 systems) or Rocky Linux 10	$0
This course and all 12 lab guides	$0
RHCSA exam (EX200)	~$500
Hands-on mock exams — linuxcert.guru	Low-cost; check the site for current pricing
CompTIA Linux+ (XK0-006)	~$369 · optional confidence builder
LFCS exam	~$395 · optional confidence builder
CompTIA Security+ — if you need DoD 8140 privileged access	~$404 · often employer-funded
CompTIA CySA+ — higher-tier work roles	~$425 · optional, usually employer-funded
Lab machines — VMs on hardware you already own	$0
Used laptop, only if your current machine can't spare the RAM	~$100–$300 one-time
Exam retake, if it comes to that	Budget for one. Most people who fail, pass the second time.

The cheapest way to fail

Booking the exam before you can finish a timed mock cleanly. The RHCSA is a single sitting with no partial credit — sitting it unprepared costs you $500 and a month of momentum. Pass a full mock with everything surviving a reboot *first*, then book.

10 Exam Day and Staying On Track

The habits that get you there, and the checklist for the morning itself

→ **Reboot after every lab.** The objectives require configurations to persist without intervention. Make rebooting a reflex now and you will never lose marks to it.

→ **Keep a break/fix log.** Every error you hit and how you fixed it, in a text file on the machine. By Week 12 it becomes your personal study guide, and it is worth more than any book.

→ **Live in the man pages.** They are the only documentation you get. Practise finding things with man -k and section numbers until it is faster than searching the web would be.

→ **Break things on purpose.** Every lab guide ends with a break/fix drill for a reason. Recovering a broken system under time pressure is the actual skill being tested.

→ **Verify, then move on.** There is no partial credit. Finishing eight tasks correctly beats starting twelve. Check each one before you leave it.

→ **Don't disable SELinux.** It is tempting when something won't work, and it will cost you the security domain outright. Learn to read a denial instead — Week 10 covers exactly that.

→ **Snapshot before every lab.** It removes the fear of breaking things, which is the single biggest thing slowing beginners down.

→ **Book the exam to create a deadline.** Once you can pass a timed mock, book it within two weeks. Open-ended study drifts.

The morning of the exam

- Government-issued photo ID ready — check the name matches your Red Hat account exactly.
- Test your machine, camera, and internet against the proctoring requirements the day *before*, not the morning of.
- Clear your desk completely. No notes, no books, no second monitor, no phone.
- Eat something. It is a long sitting and there is no meaningful break.
- First five minutes: read every task before you start any of them. Do the ones you know cold first.
- Last fifteen minutes: reboot and re-verify everything. This is not optional — it is where marks are won and lost.
- If a task will not work, leave it and come back. Do not spend forty minutes on one item and lose three easy ones.

If you don't pass the first time
It is a hard exam and plenty of good administrators fail it once. You get a score report by domain — that tells you precisely what to rebuild. Work those domains for two weeks, sit another mock, and book the retake. Most people who fail pass on the second attempt.

11 Why I Built This Guide

A note from Shea Bennett, founder of Shea's Tech

Shea Bennett started his career in Information Technology as a Helpdesk Analyst with no formal training or background in IT. He had no experience when he left the U.S. Navy, where he served as a Hospital Corpsman assigned to a Marine Corps battalion as a Fleet Marine Force (FMF) combat medic. His dream was to become a medical doctor, but the cost of medical education was simply too high — not something the Montgomery GI Bill alone could cover.

While working as a surgical technologist passing instruments at Naval Medical Center San Diego, he met engineers from a healthcare imaging company and asked one of them for help getting into IT. To his surprise, the man helped him land his first real IT job. Shea then spent the next few years training onsite at that company's headquarters in northern New Jersey, flying back and forth from San Diego. He struggled at first — he's a visual learner, and in 2007 the most common way to learn IT was in-person bootcamps or classroom training, neither of which he could afford.

Then he found video-based IT training, and never looked back. Shea is **100% self-taught**. He went on to earn a Bachelor's and a Master's degree from National University in San Diego, and trained extensively alongside various government agencies and U.S. government IT contractors. He believes anyone can learn IT on their own without paying for a bootcamp — but that mentorship and networking, in person or on live video, are critical, because IT is a team sport.

He later married a Somali-Kenyan woman who opened his eyes to a world where most people can't afford the technology and training taken for granted in the U.S. and other wealthy countries — something he saw firsthand living and traveling in Kenya and beyond.

This guide is also an homage to his own start. When Shea got out of the Navy in San Diego with no money for anything beyond a cup of instant ramen from 7-Eleven, he had to focus on what mattered and hold onto every dollar just to make it. He eventually moved back home to Philadelphia, and after a year of real struggle, found a job in the DC Metro area — and never looked back.

> *This was built for that fundi in Kenya, that hustler in Nigeria, the determined in Rwanda or Ethiopia. The hungry surviving in the UAE. The Indonesian or Filipino brother or sister — determined, but never given a chance or guidance.*
>
> *It could be that street vendor in East London, West or North Philadelphia, the hoods of NJ, or in New York City. Or that front-desk hotel worker dreaming of changing their life.*
>
> **Bismillah...**

If anything here is wrong, report it and we'll fix it.

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13 Credits and Trademarks

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Product or mark	Belongs to	Used here for
Red Hat, RHEL, Red Hat Enterprise Linux, RHCSA, RHCE, EX200	Red Hat, Inc.	The certification this course prepares you for, and the operating system it is based on. Red Hat publishes the exam objectives that this course maps to. redhat.com
Rocky Linux	Rocky Enterprise Software Foundation	A free, community-maintained rebuild of RHEL, used here as the no-account practice option. rockylinux.org
Ubuntu	Canonical Ltd.	Recommended as a host operating system for running the lab virtual machines. ubuntu.com
VirtualBox	Oracle Corporation	The default hypervisor recommendation for the home lab. virtualbox.org
VMware Workstation Pro, Fusion	Broadcom Inc.	An alternative hypervisor, free for personal use. vmware.com
Proxmox VE	Proxmox Server Solutions GmbH	Bare-metal hypervisor option for a dedicated lab machine. proxmox.com
Podman	Red Hat, Inc. and the Podman community	Container engine used by the low-RAM lab script. podman.io
CompTIA, Security+, CySA+, Linux+	CompTIA, Inc.	Named as baseline and analyst certifications for DoD 8140 work roles, and as an optional confidence builder. comptia.org
DoD 8140, DoD 8570.01-M	U.S. Department of Defense	Referenced when explaining cyber workforce qualification requirements for government contractors. public.cyber.mil
Linux Foundation, LFCS, CKA	The Linux Foundation	Named as an optional confidence-builder exam and training source. linuxfoundation.org
LPI, Linux Essentials	Linux Professional Institute	Named as an optional vendor-neutral certification path. lpi.org
killer.sh	killer.sh	Exam simulator referenced for Linux Foundation certification preparation. killer.sh
Google Cloud, AWS, DigitalOcean, Vultr	Google LLC, Amazon.com Inc., DigitalOcean Holdings Inc., The Constant Company LLC	Cloud providers supported by the hosted lab script.

Product or mark	Belongs to	Used here for
Linux	Linus Torvalds	The kernel all of this runs on, registered trademark in the US and other countries.

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